

Network Flow

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Max Flow: Problem Formulation

Input: A flow network $G = (V, E, c)$, $c : E \rightarrow \mathbb{R}^+$

$s \in V$ a source and $t \in V$ a sink

$f : E \rightarrow \mathbb{R}^+$ ($f_e = f(e)$) is a flow if it satisfies

1 **capacity constraints** $\forall e \in E : 0 \leq f_e \leq c_e$

2 **flow conservation constraints** $\forall v \in V, v \neq s, t : f^{out}(v) = f^{in}(v)$

$$size(f) = f^{out}(s) = f^{in}(t)$$

Output: A flow f of maximum size

Let f be a flow in G and let $[A, \bar{A}]$ be any $s - t$ cut in G , then

$$size(f) \leq c([A, \bar{A}])$$

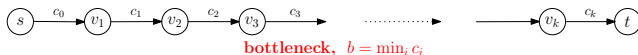
Max Flow: Adding Flow along a path

Greedy Algorithm – build up flow little bit at a time

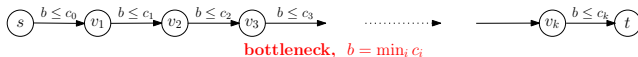
1 Start with a 0 flow

▷ Note: the flow f with $f_e = 0$ for all $e \in E$ satisfies both constraints

2 Add more flow to f via a $s - t$ path



Adding flow along a path keeps flow conservation constraints satisfied

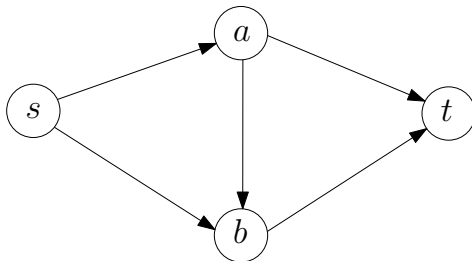


Adding flow equal to path-bottleneck keeps capacity constraints satisfied

The two ensure any intermediate flow by greedy algorithm is valid

Max Flow: Problems with the algorithm

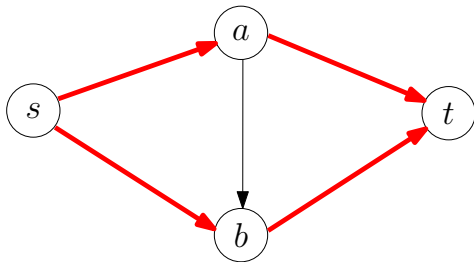
Consider the flow network – all edge capacities are 1



Max Flow: Problems with the algorithm

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The max flow clearly is of size 2

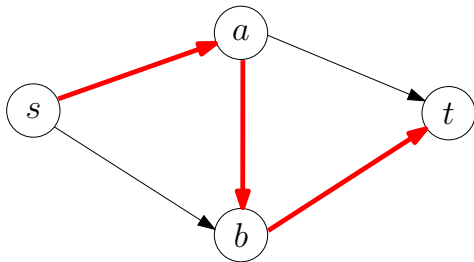


Max Flow: Problems with the algorithm

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If the greedy algorithm adds a flow of size 1 via the $s - t$ path s, a, b, t



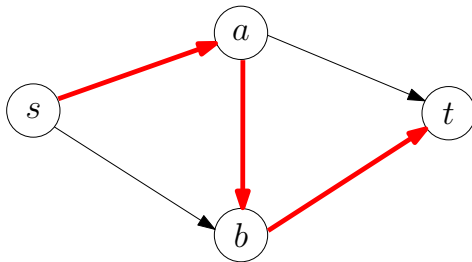
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No $s - t$ path in the remaining graph



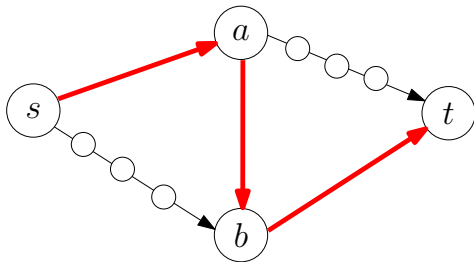
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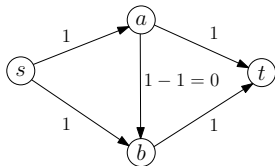
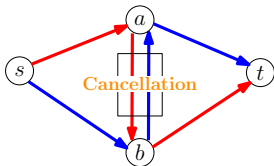
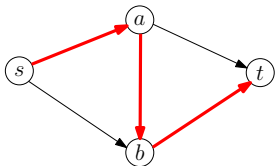


▷ The issue is not the hop-length of paths

Max Flow: Fix for the algorithm

A more general way of pushing further flow is to

- 1 Push forward flow on edges where some capacity is remaining
- 2 Cancel existing flow on edges ▷ pushing flow backward



- Add one unit of flow via the s, a, b, t path
- Add one unit of flow via the s, b, a, t path
 - ▷ $ba \notin E$, but we cancel the existing flow on edge $ab \in E$
- Add the red and blue flows

Cancellation of existing flows on edges (if need be) is the right framework to add more flow

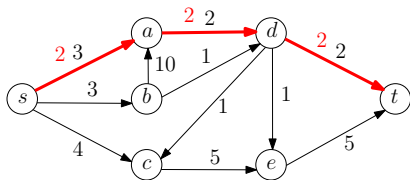
Residual network

- Provides, a systematic way to search for the right places to cancel flow and adding more flow
- Associated with a flow f , it encodes “places” where f can be increased (by adding new flows and canceling existing flows)

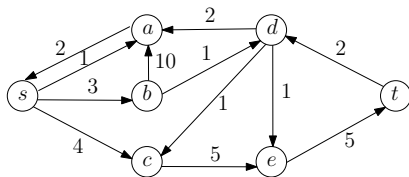
Max Flow: Residual Network

Given network G and flow f , the residual graph G_f of G w.r.t f is

- Vertex set of G_f is the same as that of G
- **Forward edges:** For each $e = uv$ of G on which $f_e < c_e$, there is an edge $e = uv$ in G_f with capacity $c_e - f_e > 0$
- **Backward edges:** For each edge $e = uv$ of G on which $f_e > 0$, there is an edge $e' = vu$ in G_f with capacity f_e



Flow network with flow shown in red



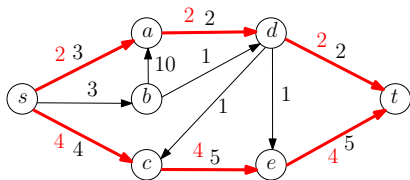
The corresponding residual network

▷ For any G and f , G_f has at most twice as many edges as G

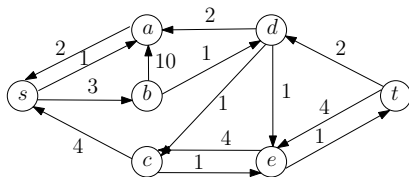
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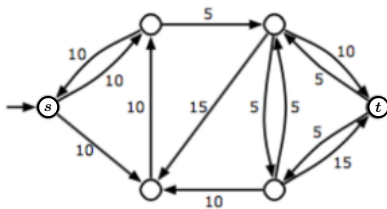
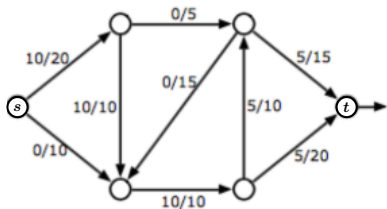
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▷ For any G and f , G_f has at most twice as many edges as G

Max Flow: Augmenting Path

An **augmenting path** is a simple $s - t$ path in the residual graph G_f

▷ It is used to augment the flow f

For an augmenting path P in G_f , $bottleneck(P, f) = \min_{e \in P} c'_e$

▷ the minimum residual capacity of any edge on P in G_f

Note c'_e is the residual capacity of the edge e (its capacity in G_f)

Algorithm $AUGMENT(P, f)$ augment flow using a path P in G_f

$b \leftarrow bottleneck(P, f)$

$f' \leftarrow f$

for each edge $e = uv \in P$ **do**

if e is a forward edge **then**

$f'_e \leftarrow f_e + b$

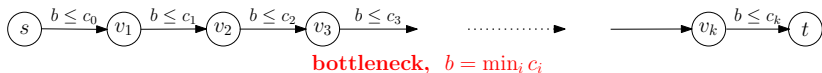
else if e is a backward edge **then**

$f'_{vu} \leftarrow f_{vu} - b$

Max Flow: Correctness of AUGMENT(P, f)

The output f' of AUGMENT(P, f) is a flow

f' satisfies capacity constraints $\forall e \in E : 0 \leq f'_e \leq c'_e$



- **Case 1:** e is not on P , $f'_e = f_e$
- **Case 2:** If $e \in P$ is a forward edge ($c'_e = c_e - f_e$), then $f'_e = f_e + b$
 - Since $0 < b \leq c'_e = c_e - f_e$
 - We have $0 \leq f_e \leq f'_e = f_e + b \leq f_e + c_e - f_e = c_e$
- **Case 3:** If $e \in P$ is a backward edge ($c'_e = f_e$), then $f'_e = f_e - b$
 - Since $0 < b \leq c'_e = f_e$
 - We have $c_e \geq f_e \geq f'_e = f_e - b \geq f_e - f_e = 0$

Max Flow: Correctness of AUGMENT(P, f)

The output f' of AUGMENT(P, f) is a flow

f' satisfies the flow conservation $\forall v \in V, v \neq s, t \quad f'^{out}(v) = f'^{in}(v)$



■ **Case 1:** e_{in} is forward edge and e_{out} is forward edge

■ $f'^{out}(v) = f^{out}(v) + b$ as $f'_{vy} = f_{vy} + b$ and

$f'^{in}(v) = f^{in}(v) + b$ as $f'_{xv} = f_{xv} + b$

■ Hence change in in-flow to v = change in out-flow from v

■ **Case 2:** e_{in} is reverse edge and e_{out} is forward edge

■ $f'^{out}(v) = f^{out}(v) - b + b$ as $f'_{vx} = f_{vx} - b$ (since vx is a reverse edge)
and $f'_{vy} = f_{vy} + b$ (since vy is a forward edge)

■ Hence $f'^{out}(v) = f'^{in}(v)$

■ **Case 3:** e_{in} is reverse edge and e_{out} is reverse edge

■ **Case 4:** e_{in} is forward edge and e_{out} is reverse edge